

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

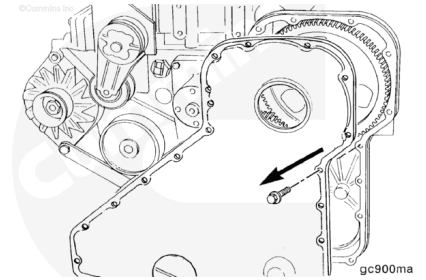


- Close the fuel supply valve. Refer to the original equipment manufacturer (OEM) instructions.
- Disconnect the batteries. Refer to the OEM service manual.
- Remove the fan drive belt. Refer to Procedure 008-002 in Section 8. (</qs3/pubsys2/xml/en/procedures/100/100-008-002.html>)
- Remove the fan pulley and spacer. Refer to Procedure 008-039 in Section 8. (</qs3/pubsys2/xml/en/procedures/100/100-008-039.html>)
- Remove the vibration damper and pulley.
 - Use the following procedure for C Series engines. Refer to Procedure 001-052 in Section 1. (</qs3/pubsys2/xml/en/procedures/41/41-001-052-tr.html>)
 - Use the following procedure for C8.3 G, C Gas Plus, and L Gas Plus engines. Refer to Procedure 001-052 in Section 1. (</qs3/pubsys2/xml/en/procedures/100/100-001-052.html>)
- Remove the crankshaft oil seal dust shield, if used.

Remove

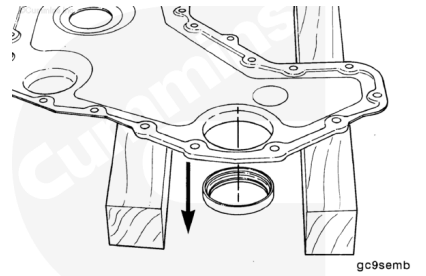
Remove the front gear cover mounting capscrews.

Remove the front gear cover.



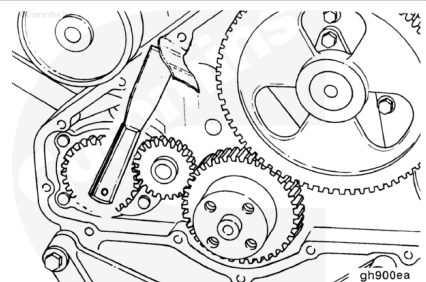
Disassemble

Remove the crankshaft seal. Refer to Procedure 001-023 in Section 1. (</qs3/pubsys2/xml/en/procedures/100/100-001-023.html>)



Clean and Inspect for Reuse

Clean the front gear housing gasket sealing surface.

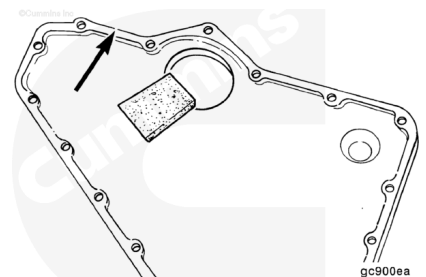


⚠ CAUTION ⚠

The seal lip and sealing surface on the crankshaft must be clean and free from all oil residue to prevent seal leaks.

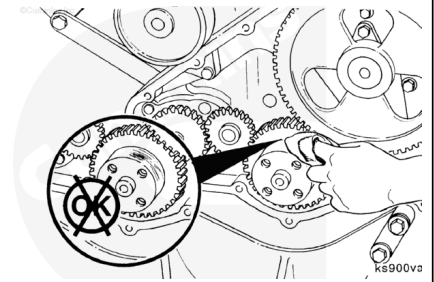
Clean the gasket sealing surfaces and the surface between the oil seal and gear cover.

Thoroughly clean the seal contact area of the crankshaft.



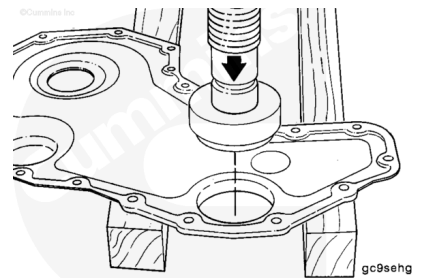
Inspect the gear cover for cracks, dents, and other damage.

Inspect the sealing surface of the crankshaft for wear. If the crankshaft has a groove due to the oil seal, a wear sleeve **must** be used to prevent seal leakage.



Assemble

Install a new crankshaft seal. Refer to Procedure 001-023 in Section 1. (</qs3/pubsys2/xml/en/procedures/100/100-001-023.html>)



Install

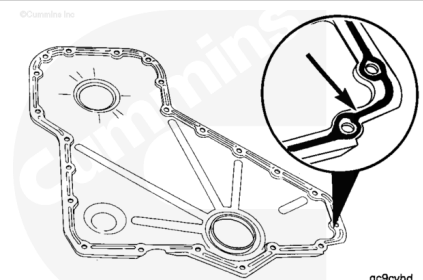
Note : Do not remove the plastic seal pilot tool from the lubricating oil seal at this time. Use the plastic seal pilot tool to guide the seal on the crankshaft.

Note : Install three guide pins, Part Number 3164977, to improve alignment of the front cover and front seal to the gear housing and crankshaft.

Apply a 4-mm wide bead of ThreeBond, Part Number 3823494, along the entire flange of the gear cover.

Follow the grooves in the flange; circle all of the capscrew holes with ThreeBond, Part Number 3823494. Do **not** use a paper gasket with the gear cover.

Note : The front cover **must** be installed and the capscrews tightened within 15 minutes from when the ThreeBond was applied. If the ThreeBond, Part Number 3823494, appears skinned over, it will not create a good seal. It **must** be cleaned from the front cover and new ThreeBond, Part Number 3823494, applied.



Install the front gear cover and tighten the capscrews.

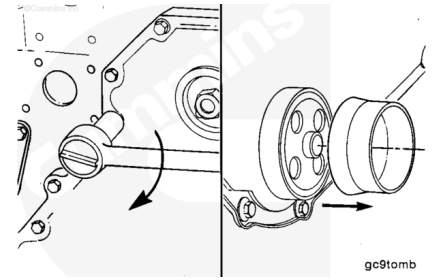
Tighten the long cover capscrews with 11-mm heads to:

Torque Value: 40 n•m [30 ft-lb]

Tighten the short cover capscrews with the 10-mm heads to:

Torque Value: 30 n•m [22 ft-lb]

Remove the plastic pilot tool from the crankshaft.



Finishing Steps

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Install the crankshaft oil seal dust shield, if used, as follows:

- Push the dust shield onto the crankshaft with the part number facing out away from the engine.
- The dust shield **must** contact the oil seal.

Note : No lubrication is required for installation.

- Install the vibration damper and pulley.
 - Use the following procedure for C Series engines. Refer to Procedure 001-052 in Section 1.
(/qs3/pubsys2/xml/en/procedures/41/41-001-052-tr.html)
 - Use the following procedure for C8.3 G, C Gas Plus, and L Gas Plus engines. Refer to Procedure 001-052 in Section 1.
(/qs3/pubsys2/xml/en/procedures/100/100-001-052.html)
- Install the fan pulley and spacer. Refer to Procedure 008-039 in Section 8.
(/qs3/pubsys2/xml/en/procedures/100/100-008-039.html)
- Install the fan drive belt. Refer to Procedure 008-002 in Section 8. (/qs3/pubsys2/xml/en/procedures/100/100-008-002.html)

- Connect the batteries. Refer to the OEM service manual.
- Open the fuel supply valve. Refer to the OEM instructions.
- Operate the engine 5 to 10 minutes to check for leaks and proper operation.

Last Modified: 26-Aug-2014
